

1. TISSUE PROCESSING UNIT

Processing of fixed tissue through graded alcohols, clearing agent and paraffin infiltration before embedding.

SOP No.: LAB-HISTO-01	Revision: 01	Effective: _____
Department: Histopathology / Research	Prepared by: _____	Approved by: _____

Purpose

- To remove water from fixed tissue, clear the tissue and infiltrate it with molten paraffin wax to produce a sectionable paraffin block.

PPE & Safety

- Wear lab coat, gloves and eye protection. Handle formalin, alcohols and clearing agents in a ventilated area. Treat all specimens as potentially infectious. Keep the unit away from ignition sources where flammable reagents are used.

Materials

- Fixed tissue cassettes; labelled processing baskets; graded alcohols; clearing reagent (e.g., xylene or approved substitute); paraffin wax; forceps; waste containers.

Procedure

- Verify instrument cleanliness, reagent levels, program and temperature settings before starting.
- Load correctly labelled, securely closed cassettes into the basket. Select the validated tissue-processing program appropriate for tissue type and cassette load.
- Typical sequence: fixation/hold (as applicable) → graded alcohol dehydration → clearing → paraffin infiltration. Use the laboratory's validated times and temperatures; do not substitute unvalidated schedules.
- Start the cycle and monitor for alarms, leaks, abnormal temperature or reagent depletion.
- At completion, remove the basket carefully and proceed promptly to the embedding station.
- Record program/cycle details and any deviations in the equipment or specimen log.

Quality Control

- Check that tissue is adequately processed: firm but not brittle, uniformly infiltrated and free of residual water/clearing-agent problems. Poor processing may cause soft, crumbly or poorly ribboning blocks.

Shutdown & Cleaning

- Follow manufacturer instructions for standby/shutdown. Replace contaminated or exhausted reagents according to laboratory schedule. Clean spills immediately using the approved procedure. Never mix incompatible chemicals.

Troubleshooting

- Under-processing → review reagent strength, sequence and time. Over-hard/brittle tissue → review dehydration/clearing exposure. Poor infiltration → verify paraffin temperature, level and infiltration time. Persistent faults → stop use and inform the supervisor/biomedical service.

Important: This is a general SOP template. Before operational use, insert the exact manufacturer model, validated temperatures/times, reagent names, decontamination procedure, QC acceptance criteria and institutional biosafety/chemical-safety requirements.